

Anti-Gravity Treadmills Are Effective in Reducing Knee Forces

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ABSTRACT: Lower body positive pressure (LBPP) treadmills permit significant unweighting of patients and have the potential to enhance recovery following lower limb surgery. We determined the efficacy of an LBPP treadmill in reducing knee forces in vivo. Subjects, implanted with custom electronic tibial prostheses to measure forces in the knee, were tested on a treadmill housed within a LBPP chamber. Tibiofemoral forces were monitored at treadmill speeds from 1.5 mph (0.67 m/s) to 4.5 mph (2.01 m/s), treadmill incline from -10° to $+10^\circ$, and four treadmill chamber pressure settings adjusted to decrease net treadmill reaction force from 100% to 25% of the subject's body weight (BW). The peak axial tibiofemoral force ranged from 5.1 times BW at a treadmill speed of 4.5 mph (2.01 m/s) and a pressure setting of 100% BW to 0.8 times BW at 1.5 mph (0.67 m/s) and a pressure setting of 25% BW. Peak knee forces were significantly correlated with walking speed and treadmill reaction force ($R^2 = 0.77$, $p = 0.04$). The LBPP treadmill might be an effective tool in the rehabilitation of patients following lower-extremity surgery. The strong correlation between tibiofemoral force and walking speed and treadmill reaction forces allows for more precisely achieving the target knee forces desired during early rehabilitation. © 2012 Orthopaedic Research Society. Published by Wiley Periodicals, Inc. J Orthop Res 31:672–679, 2013

Keywords: total knee arthroplasty; anti-gravity treadmill; knee forces; knee biomechanics; rehabilitation; exercise

Postoperatively, an organized rehabilitation program allows for building of muscle strength and coordination, which may have been compromised prior to surgery due to painful restriction because of the underlying disease. In lower limb surgeries, a period of restricted weight bearing and immobilization is often advised to allow for healing of the surgical site. However, postoperative inactivity results in muscular atrophy and joint stiffness and increases the risk of vascular stasis and thrombosis. An effective rehabilitation prevents these complications while protecting the surgical site.

Knowledge of knee forces is valuable to evaluate the design of effective rehabilitation programs that prevent postoperative complications without jeopardizing the repair process surgical site. Weight bearing activities such as walking are often considered highly effective in rehabilitation and return to function. However, due to the high knee forces (between two and three times body weight), non-weight bearing, or partial weight bearing is often recommended, which delays return to full function. One approach is to reduce the forces at the knee during weight bearing exercise. Examples of such attempts include hydrotherapy (walking in water),¹ the use of harness systems that physically lift the subject,² the use of specialized robots in lifting and providing assistance during walking (<http://www.hstartech.com>), and lower body positive pressure chambers.³ Lower body positive pressure (LBPP) treadmills produce a significant unweighting of the subject with minimal alteration of gait kinematics. This modality has the potential to enhance recovery in patients following lower limb surgery for ACL reconstruction and meniscectomy.³

We previously measured knee forces during activities of daily living and exercise following total knee arthroplasty with tibial trays instrumented with force sensors and a telemetry system.^{4,5} Peak tibial forces varied between 0.8 times body weight (BW) while rowing to 4.5 times BW while jogging.⁶ These data are now being used to design more appropriate postoperative protocols and to advise patients regarding safe and unsafe activities. Our present objective was to determine the efficacy of an LBPP treadmill in reducing knee forces as a first step toward validating usage for a rehabilitation tool following lower limb surgery. A secondary objective was to correlate the knee forces with treadmill parameters such as pressure, speed, and incline.

METHODS

After appropriate institutional review board approval and consent, four subjects (Table 1) implanted with custom electronic tibial prostheses were tested in the LBPP chamber (Fig. 1). The inclusion criteria were all subjects implanted with the instrumented knee prosthesis. The exclusion criteria were significant pain, deformity, inability to walk at the prescribed speeds, and any condition precluding the use of positive pressure to the lower body. The electronic implants measure and wirelessly transmit the forces in the knee. A six component sensor located within the stem of the tibial tray monitors the net tibiofemoral contact force in real time. Details of the components, the accuracy, and the in vivo measured forces were previously reported.^{4,5,7,8}

Tibial forces were monitored at treadmill speeds ranging from 1.5 mph (0.67 m/s) to 4.5 mph (2.01 m/s) in 1 mph (0.45 m/s) intervals, changes in treadmill incline (up to $\pm 10^\circ$), and four treadmill chamber pressure settings (Table 2). The subjects were instructed not to use handrails, and all gait cycles processed for analysis were without the use of handrails. The treadmill pressure setting was initially calibrated to get a measure of the pressure required to completely unload the force on the treadmill (0% BW) during a static two-legged

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Table 1. Demographic Data of the Four Subjects Implanted With Instrumented Devices

Subject	Sex	Age (Years)	Weight (kg)	Height (m)
1	Male	80	67	1.64
2	Male	83	74	1.80
3	Male	79	80	1.73
4	Female	67	89	1.63

standing. Subsequent adjustments to pressure settings were made during dynamic walking. Data were collected after the walking cadence at each treadmill speed and pressure was stabilized. The treadmill pressure was adjusted for each subject to decrease net treadmill reaction force from 100% to 25% of the subject's BW. These levels were chosen to simulate the corresponding weight bearing scenarios during postoperative recovery, with 25% BW reflecting non-weight bearing, 50% reflecting toe-touch walking, 75% reflecting partial weight bearing, and 100% reflecting full weight bearing. The complete list of treadmill conditions that we evaluated are listed in Table 2. To minimize bias due to test sequence and fatigue, measurement of knee forces were repeated on different days, and treadmill speeds, incline, and pressure (Table 2), were set in a random sequence. Range of motion at the knee was recorded with an electrogoniometer. Ground reaction force was measured via force plates in the treadmill. Tibiofemoral forces were transmitted from the prostheses via the implanted telemetry system.

Data Processing

The time duration of each gait cycle was normalized to 100% for comparison among conditions and across subjects. Stride

Table 2. Treadmill Conditions Tested for Each Subject

Treadmill Speed		Treadmill Reaction Force (% Body Weight)	Treadmill Incline
Miles/h	m/s		
1.5	0.67	100%	+10°
2.5	1.12	75%	+5°
3.5	1.56	50%	0°
4.5	2.01	25%	-5°
			-10°

Four treadmill speeds were tested at four different treadmill reaction forces (16 trials). Five treadmill incline settings were tested for four different treadmill reaction forces (20 trials) at one treadmill speed (2 mph).

duration was measured as the time interval between two subsequent ipsilateral foot touch downs. Stride length was calculated from average stride duration and treadmill speed and was normalized to length of the lower leg (from greater trochanter to the ground). Cadence was defined as the number of steps per second. Axial load, anterior shear, and flexion moment were averaged from at least 10 cycles for each combination of conditions. To normalize for differences in dynamic behavior, we computed the Froude number. The Froude number is a ratio of the square of the speed of walking to the length of the lower limb and acceleration due to gravity (Equation 1)

$$Fr = \frac{S^2}{gL}$$

1

Handwritten notes: $\gamma = \omega r = 2\pi s$, $\omega = 2\pi \times 1$, $s = \sqrt{F/g}$, $s =$

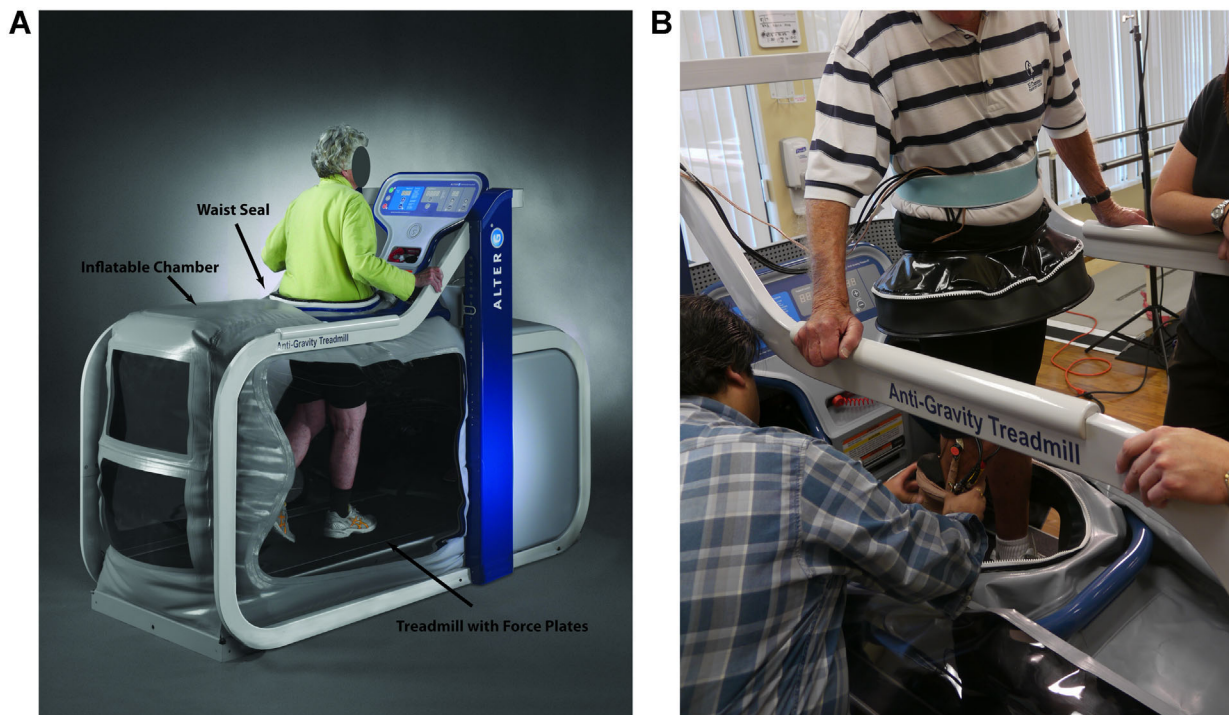


Figure 1. (A) Photograph of a lower body positive pressure treadmill demonstrating the inflatable chamber and waist seal. (B) Photograph of a subject with an electronic tibial tray wearing the waist seal and being prepared for in vivo force measurement.

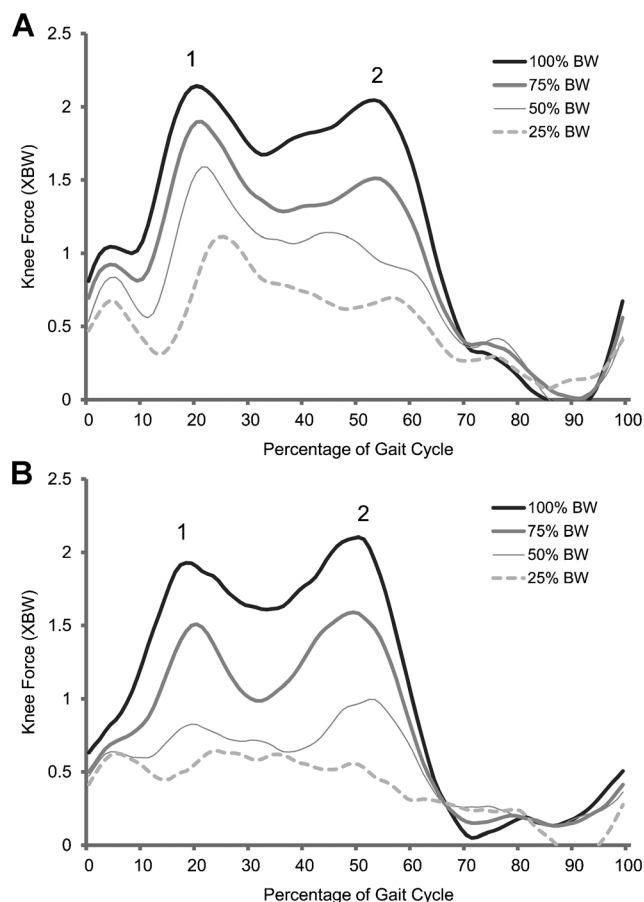


Figure 2. Representative gait cycles from two subjects (A and B) showing the consistent reduction of knee forces with increase in chamber pressure. 1 = first major peak after heel strike; 2 = second major peak before toe-off.

where Fr is the Froude number, S = treadmill speed, g = acceleration due to gravity, and L = length of lower limb.

Statistical Analysis

Repeated measures ANOVA was used to test the significance of differences in forces among the various body weights. Multiple linear regression was used to determine the strength of correlation between treadmill speed and body weight, and peak tibial force.

RESULTS

Representative gait cycles from two subjects demonstrate the consistent reduction in knee forces during walking with the reduction in treadmill reaction force due to increase in chamber pressure (Fig. 2). The peak axial knee force during treadmill use ranged from 5.1 times BW at a treadmill speed of 4.5 mph (2.01 m/s) and a pressure setting of 100% BW to 0.8 times BW at 1.5 mph (0.67 m/s) and a pressure setting of 25% BW (Fig. 3, Table 3A). The decrease in peak knee force (averaged across subjects) with decreasing ground reaction forces was consistent across all treadmill speeds (Fig. 3, $p < 0.001$). Peak knee forces were significantly correlated with walking speed and treadmill reaction force ($R^2 = 0.77$, $p = 0.04$)

$$F = -0.3 + 0.3S + 1.89BW$$

2

where F = peak knee force, S = treadmill speed (mph), BW = reaction force at treadmill as a fraction of body weight.

There was a trend toward higher peak knee forces when walking up steeper inclines, but this trend was not significant (Fig. 4, Table 3B, $p = 0.18$). Peak anterior shear did not vary with body weight setting ($p = 0.11$) or speed ($p = 0.8$); however, the change with incline approached significance ($p = 0.07$).

Effect of Treadmill Reaction Force and Speed on Peak Tibial Forces

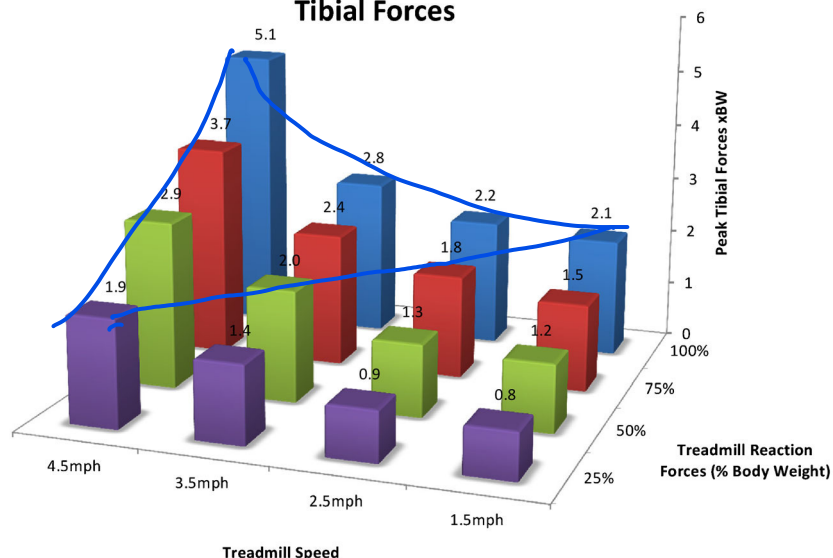


Figure 3. Peak axial knee forces were averaged across the multiple trials for each test condition, then averaged across the four subjects (at 0° incline). Axial knee forces increased with increasing treadmill speed and decreased with increasing chamber pressure (corresponding to lower net body weight).

Table 3. Tables Showing Data Variability Between Subjects and Test ConditionsA: Average (+standard deviation) of peak axial knee forces ($\times$ BW) at various treadmill reaction forces and speeds

Treadmill Reaction Force (% Body Weight)	1.5 mph (0.67 m/s)	2.5 mph (1.12 m/s)	3.5 mph (1.56 m/s)	4.5 mph (2.01 m/s)
100%	2.10 (+0.36)	2.25 (+0.23)	2.80 (+0.51)	5.05 (+0.44)
75%	1.53 (+0.19)	1.82 (+0.20)	2.36 (+0.43)	3.74 (+0.31)
50%	1.19 (+0.18)	1.27 (+0.31)	1.96 (+0.49)	2.95 (+0.05)
25%	0.82 (+0.15)	0.89 (+0.30)	1.37 (+0.23)	1.89 (+0.52)

B: Average (+standard deviation) of peak axial knee forces ($\times$ BW) at various treadmill inclines

Treadmill Reaction Force (% Body Weight)	+10°	+5°	0°	-5°	-10°
100%	2.33 (+0.42)	2.26 (+0.26)	2.10 (+0.19)	2.09 (+0.21)	2.06 (+0.27)
75%	1.87 (+0.17)	1.73 (+0.11)	1.53 (+0.18)	1.71 (+0.12)	1.58 (+0.29)
50%	1.34 (+0.23)	1.29 (+0.18)	1.19 (+0.20)	1.32 (+0.21)	1.24 (+0.28)
25%	1.13 (+0.29)	0.93 (+0.32)	0.82 (+0.29)	0.95 (+0.28)	0.95 (+0.25)

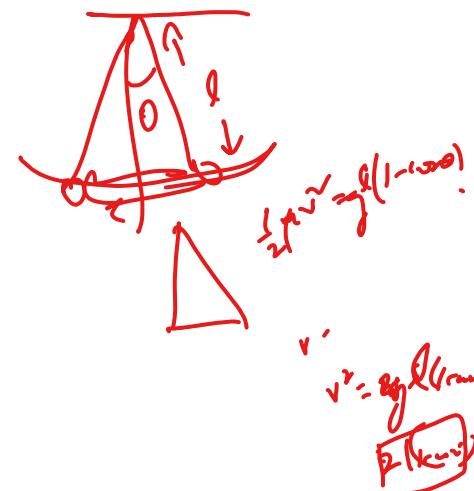
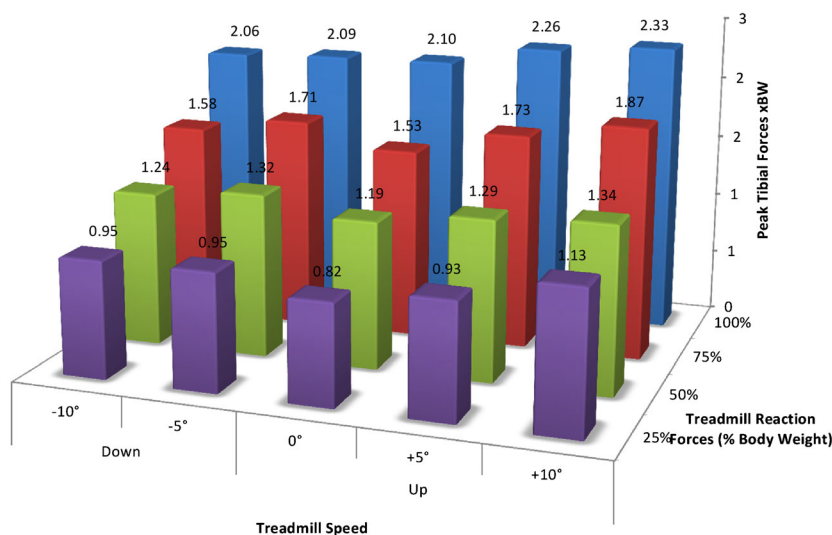
C: Average (+standard deviation) normalized walking stride length at various treadmill speeds

Treadmill Reaction Force (% Body Weight)	1.5 mph (0.67 m/s)	2.5 mph (1.12 m/s)	3.5 mph (1.56 m/s)
100%	1.03 (+0.07)	1.41 (+0.11)	1.73 (+0.12)
75%	1.04 (+0.08)	1.43 (+0.10)	1.74 (+0.09)
50%	1.07 (+0.09)	1.43 (+0.10)	1.73 (+0.09)
25%	1.04 (+0.09)	1.48 (+0.11)	1.78 (+0.08)

While average normalized stride length increased with increasing treadmill speeds, there were no significant changes with reduction in BW (Fig. 5, Table 3C). A strong linear correlation was noted with treadmill speed and cadence (Fig. 6, $p < 0.001$). To compare differences in dynamic behavior among subjects, we used the Froude number (Equation 1), which is a dimensionless speed relating the dynamics of human walking to that of a pendulum.⁹ A Froude

number of 0.25 corresponds to comfortable walking. Most normal individuals spontaneously transition from walking to running at a Froude number of 0.5, while a Froude number of 1 is the theoretical maximum.¹⁰⁻¹² The Froude numbers for each combination of treadmill speed and body weight reduction are listed in Table 4. By varying speed and body weight settings, we were able to generate a wide range of Froude numbers.

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Effect of Treadmill Incline on Peak Tibial Forces**Figure 4.** Peak knee forces did not change with treadmill incline.

Effect of Treadmill Speed on Stride Length

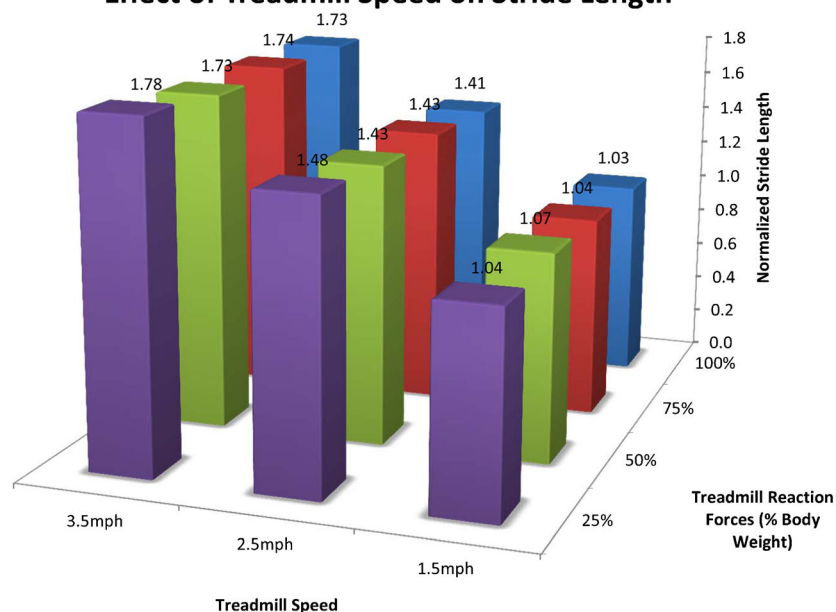


Figure 5. Average normalized stride length increased with increasing treadmill speeds.

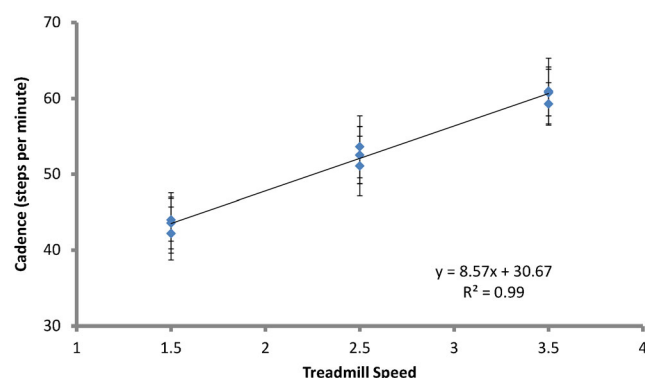


Figure 6. Cadence was linearly correlated with treadmill speed.

DISCUSSION

Early return to full range of motion, normal strength, and function is important for the success of most orthopedic procedures. Lower extremity surgeries, especially on the knee, usually involve a protracted recovery period primarily to protect the healing tissues. In surgeries for replacement of articular cartilage, such as mosaicplasty and allograft/autograft, a graduated increase in weight bearing is advised as

graft interface healing progresses. Patients undergoing autologous chondrocyte implantation are at risk for surgical failure if load bearing is excessive during the early postoperative period.¹³ We previously reported on the initial findings of reduced knee forces in an LBPP chamber in a pilot study involving one subject.^{14,15} This report represents the first systematic analysis of BW reduction, treadmill speed, and treadmill incline on knee forces to validate the use of LBPP in reducing forces at the knee.

Peak axial force at the tibial tray was consistently reduced at lower body weight settings, generating forces at the knee ranging from <1 to >5 times BW. As we previously reported for regular treadmill walking, walking speed correlated directly with tibiofemoral forces.⁶ Normalizing knee forces to the body weight of each patient reduced the between-subject variability. This enabled us to generate an equation using multiple linear regression that predicted peak axial tibiofemoral forces from treadmill speed and body weight setting with an R^2 of 0.77.

At zero pressure (100% BW treadmill reaction force) knee forces peaked between 2 and 3 times BW at treadmill speeds ranging from 1.5 to 3.5 mph. These results were similar to knee forces we previously reported for treadmill walking or walking on level

Table 4. Average (+Standard Deviation) Froude Number by Treadmill Speed and Body Weight

Treadmill Speed		Treadmill Reaction Force (% Body Weight)			
mph	m/s	100%	75%	50%	25%
1.5	0.67056	0.05 (+0.002)	0.07 (+0.003)	0.10 (+0.004)	0.21 (+0.008)
2.5	1.1176	0.14 (+0.006)	0.19 (+0.007)	0.29 (+0.011)	0.57 (+0.022)
3.5	1.56464	0.28 (+0.011)	0.37 (+0.015)	0.56 (+0.022)	1.12 (+0.044)

Table 5. Knee Forces During Activities of Daily Living & Exercise (Adapted From D'Lima et al., J Eng Med, 2012)

Activity	Peak Tibial Forces ($\times$ BW)	Notes
Rowing machine	0.9 ± 0.1	
Stationary bicycling	1.0–1.5	Level 1–5; 60–90 rpm
Knee extension	1.5 ± 0.0	Resistance = $0.2 \times$ BW
Treadmill walking	2.1 ± 0.2	1–3 miles/h
Walking ^a	1.3 ± 0.2	2.5 miles/h on AlterG treadmill at 50% BW pressure
Elliptical level 1	2.3 ± 0.2	
Elliptical level 11	2.2 ± 0.3	
StairMaster level 1	2.4 ± 0.1	StairMaster level 1
StairMaster level 3	3.3 ± 0.3	StairMaster level 3
Power walking	2.8 ± 0.4	4 miles/h on treadmill
Jogging	3.1–3.6	Tumor replacement prosthesis
Jogging	4.2 ± 0.2	5 miles/h on treadmill
Jogging ^a	2.9 ± 0.1	4.5 miles/h on AlterG treadmill at 50% BW pressure
Golf (trailing knee)	3.0 ± 0.2	Right knee for a right handed golfer
Golf (lead knee)	4.4 ± 0.1	Left knee for a right handed golfer
Tennis backhand	3.5 ± 0.6	
Tennis serve	4.2 ± 0.1	
Tennis forehand	4.3 ± 0.4	

^aResults from present study.

ground, which indicated that baseline measurements were not affected by the pressure chamber. To place these peak knee forces in the context of other activities, we compared these results with our previous report on forces measured during rehabilitation, exercise, and sporting activities (Table 5). Knee forces fell within the low end of those measured, along with low impact activities such as rowing and stationary bicycling.

Exercising in water (hydrotherapy) has been recommended as an effective rehabilitation method. The buoyancy created while walking in water can decrease ground reaction forces based on the portion of the body submerged. For example submerging in water up to the level of the axilla reduces ground reaction forces by 80%.¹⁶ However, the increased resistance to motion also alters hip, knee, and ankle moments.^{14,16,17} Despite reduced ground reaction forces, joint moments can increase due to the increased resistance in water.¹⁸ The coupled effect of buoyancy and hydrodynamic drag also affect inter-joint coordination.¹⁸ One advantage of the LBPP chamber is that it does not affect hydrodynamic and aerodynamic drag.

Lower limb exercise in water can be used to exercise the muscles of the leg over a full range of motion at slow speeds. However, the hydrodynamic drag makes it difficult to conduct high-frequency cyclic activities such as jogging or running. Increasing walking speed results in a significant increase in joint moments, which can increase joint forces.¹⁶ Our data also indicate that knee forces can be reduced to <2 times BW while jogging at 4.5 mph (2.01 m/s) in the LBPP chamber. This permits exercising at running speeds while maintaining knee forces below those of comfortable walking.

The magnitude of the anterior component of ground reaction force is similar while walking on land or in water at a comfortable speed.¹⁶ However, increasing the speed of walking in water substantially increased the anteroposterior component of ground reaction forces, which may result increase in shear at the knee joint.¹⁶ In our study, we directly measured anteroposterior shear at the tibial tray and found that it correlated directly with the reduction in overall tibial forces. The only treadmill parameter that affected anterior shear (other than the reduction in ground reaction force) was the incline. This feature of an LBPP treadmill to independently vary ground reaction force, treadmill speed, and treadmill inclines, can therefore be valuable in controlling forces at the knee more precisely, for example, while protecting a knee after anterior cruciate ligament reconstruction.

Alternative methods of decreasing the ground reaction forces are the use of body weight suspension or specialized harness systems, which physically lift the subject off the exercise surface.^{2,19} However, the effect of these methods for reducing body weight on knee forces has not been studied. Analysis of the spatiotemporal changes in the kinematics and kinetics of lower limb joints at different levels of body weight suspensions was reported. Body weight support resulted in a longer stride and decreased cadence.²⁰ In our study, while normalized stride length and cadence changed with treadmill speed, no significant changes were found with reduction in body weight due to increased pressure.

One challenge when comparing across subjects is to account for between-subject variability in kinetics. The common method is to normalize by body weight (for

SW x Height

forces) and the product of body weight and height (for moments). This method is effective in minimizing the effect of height and weight on joint forces. However, while comparing kinetics and kinematics among differing individuals at different speeds and gravity settings, one should normalize for differences in dynamic behavior. One popular method is the Froude number, which relates the dynamics of human walking to that of a pendulum.⁹ A Froude number of 0.25 is considered the dimensionless speed corresponding to optimal walking. By varying treadmill speed and ground reaction force (gravity), we were able to test the subjects at average Froude numbers varying from 0.05 to 1.12. At chamber pressures generating ground reaction forces of 25% BW, the Froude number approached one (the theoretical limit of walking). At the lowest body weight setting (25%), the calculated Froude number was higher than the theoretical maximum of one. This is because the treadmill was moving faster than the propulsion provided from the lower legs. Using a standard treadmill, the subject would not have been able to stay in place without the support of the chambers seal around the waist. We chose to keep the defined treadmill speeds constant between subjects rather than keep Froude numbers constant. However, the variation in Froude number among subjects was low. For example, at a treadmill speed of 3.5 mph (1.56 m/s), the range of Froude numbers was only 0.25–0.27. With a given percentage of BW, one can adjust the treadmill speed to each individual to equate a Froude number of 0.25, which may be a useful method of keeping patients exercising at an optimal speed of walking.

The limitations of our study are the small sample size and the fact that these patients are not representative of the typical patients undergoing osteochondral repair or anterior cruciate ligament reconstruction that may benefit from force reduction during rehabilitation. In addition, our patients are older, have total knee replacement implants, and do not have an intact anterior cruciate ligament. Aging increases the variability in lower limb kinematics and kinetics during treadmill walking with lower body suspension systems.²¹ Therefore, the absolute values of knee kinetics and kinematics may not directly extrapolate to other populations. In young (average 23 years of age) healthy volunteers, a strong linear correlation was noted with treadmill speed and step frequency (cadence).²² In our data, we noted a similar linear correlation. While our sample size was smaller ($N = 4$, compared to $N = 12$) and the linear coefficients were different (reflecting the difference in age and surgical status), this correlation was maintained across all pressure settings (25% to 100% BW). While knee force magnitudes are significantly altered by modulating the ground reaction forces, other kinematic parameters such as cadence, stride length, and stride duration were relatively unaffected. This supports the feasibility of designing a graded rehabilitation program with gradual restoration to full weight

bearing without significantly confounding kinematic behavior.

In conclusion, we demonstrated the efficacy of using the lower body positive pressure treadmill. The LBPP treadmill allowed for a controlled decrease of the joint forces and might be an effective tool in the rehabilitation of patients following lower-extremity surgery. While we measured forces in subjects with total knee implants, a relative reduction in knee forces with reduction in body weight due to increased chamber pressure is also likely in patients who undergo other types of lower-extremity surgery.

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